

LUCY C. MURRAY

Contact Information

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Professional Summary

- Senior Data Science and Analytics leader with 7+ years of experience spearheading advanced machine learning models, statistical analysis, and enterprise data engineering pipelines across complex sectors.
- Proven track record designing robust predictive algorithms and migrating legacy architectures to modern, scalable production frameworks using Python, SQL, and cloud-integrated data environments.
- Expert at translating intricate technical patterns and AI data solutions into high-visibility visualizations and actionable strategic insights for cross-functional business stakeholders and non-technical leadership.
- Accomplished technical mentor who has scaled organizational data literacy and guided engineering teams through rigorous code reviews, MLOps best practices, and Agile delivery.

Experience

TECHNOMICS, INC. | Senior Analyst | August 2021 – Present

Project Management & Leadership

- Lead advanced data science, predictive modeling, and analytics initiatives to develop scalable algorithms and machine learning solutions for federal, defense, and international enterprise environments.
- Manage cross-functional engineering teams of up to 14+ members under Agile/Kanban frameworks, directing sprint agendas, conducting technical code reviews, and mentoring junior analysts in Python and statistical techniques.

Database Management & Advanced Analytics

- Designed and deployed a machine learning prioritization model within production environments leveraging Python, SQL, and Palantir Foundry pipelines, reducing process subjectivity and optimizing cloud resource allocation.
- Championed enterprise database migrations from legacy structures to modern, high-performance architectures using DuckDB, ensuring strict data governance, clean schema transformations, and rigorous feature engineering.
- Replaced manual, spreadsheet-heavy workflows with automated, highly optimized programmatic pipelines to discover meaningful patterns, trends, and statistical relationships in massive data tranches.
- Engineered interactive, high-visibility business intelligence tools and data visualizations, translating complex risk metrics and algorithmic outcomes into clear recommendations for executive stakeholders.

KCIC, LLC | Senior Data Analyst | March 2019 – August 2021

Database Engineering & Quality Control

- Automated data workflows and predictive solutions, developing advanced SQL views, stored procedures, and indexes to optimize data retrieval speeds and front-end infrastructure performance.
- Leveraged Python and SQL to clean, transform, and analyze massive relational datasets, identifying key operational trends and generating foundational industry analytics reports.
- Designed and implemented automated data validation routines and causal analysis to audit legacy records under strict 48-hour data ingestion and production deployment deadlines.

Project Management & Professional Training

- Designed and executed firm-wide data analysis training programs that standardized database architecture ingestion, reporting protocols, and analytics adoption across multiple cross-functional teams.

Education

- Professional Certificate in Data Science, Montgomery College | 2019 – 2020
- B.A., Economics, Government & Politics, University of Maryland, College Park | 2013 – 2017

Core Competencies

Leadership: Agile Methodologies (Kanban), Project Management, Cross-functional Team Leadership, Mentorship & Code Reviews

Data Science & Analytics: Predictive Prioritization Modeling, Factor Analysis, Statistical Sampling, Data Visualization, MLOps

Technical Toolkit: Python, R (Shiny, ggplot2), SQL (MS SQL, DuckDB), Palantir Foundry, Power BI, Git/GitLab